Prodigy Anesthesia Quick Review

Cardiac Surgery

During cardiopulmonary bypass, what are some techniques that offer cerebral protection?	Although many techniques have been introduced into practice to combat post cardiopulmonary bypass neurologic dysfunction, only hypothermia has been proven to be efficacious. *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 521.*
At what point will a patient with atherosclerotic coronary disease exhibit symptoms?	Coronary vessels that become 70% occluded will produce symptoms due to a reduction in blood flow to the coronaries (stable angina). *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 1-2.*
What are the two major determinants of coronary artery perfusion pressure?	Left ventricular end-diastolic pressure and aortic diastolic pressure. *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 510.*
What common hemodynamic alteration has the highest likelihood to cause myocardial ischemia?	Tachycardia decreases the myocardial oxygen supply while increasing the demand and has the highest likelihood of resulting in myocardial ischemia. *Miller RD, Pardo MC. Basics of Anesthesia. 6th ed. Philadelphia: Elsevier Saunders; 2011: 641.*
When the balloon of a PA catheter is not inflated, what pressures are reflected during systole and diastole?	Right ventricular systolic pressure is transduced during systole and PA diastolic pressure is transduced during diastole. The PA diastolic pressure is also a representation of left atrial (LA) pressure. *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 300.*
List complications that can arise from the placement of PA catheters.	Dysrhythmias, knotting of the catheter, injury to cardiac valves, and rupture of the pulmonary artery. *Miller RD, Pardo MC. Basics of Anesthesia. 6th ed. Philadelphia: Elsevier Saunders; 2011: 324.*
What patient populations should receive pulmonary artery (PA) catheters during CABG?	Patients who are undergoing complex procedures, when large volume shifts are anticipated, or the patient exhibits pulmonary hypertension, diastolic dysfunction, left ventricular failure, or right ventricular failure. *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 525.*
What cross allergy is of concern with a DM patient on NPH insulin presenting for cardiac surgery?	Protamine *Miller RD, Pardo MC. Basics of Anesthesia. 6th ed. Philadelphia: Elsevier Saunders; 2011: 412.*

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Why are patients with diabetes mellitus at an increased risk for adverse outcomes during and after cardiac surgery?	The possibility of silent ischemia and MI places patients with diabetes mellitus at an increased risk *Longnecker DE, Newman MF, Brown DL, Zapol WM. Anesthesiology. 2nd ed. New York: McGraw-Hill; 2012: 904.*
How is cardioplegia accomplished during cardiac surgery?	By the administration of a potassium solution into the coronary circulation *Miller RD, Pardo MC. Basics of Anesthesia. 6th ed. Philadelphia: Elsevier Saunders; 2011: 409.*
An EF less than 30% is indicative of what?	Poor cardiac function. *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 514.*
When the balloon of a PA catheter is inflated, what pressures can be obtained?	During ventricular systole, the left atrial filling pressure is reflected, upon ventricular diastole, the left ventricular filling pressure is measured. When the balloon is inflated, it is termed "wedged". *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 300.*
How does nitroglycerin treat myocardial ischemia?	Nitroglycerin relieves myocardial ischemia primarily through coronary vasodilation and reduction of preload reduction from venodilation. *Longnecker DE, Newman MF, Brown DL, Zapol WM. Anesthesiology. 2nd ed. New York: McGraw-Hill; 2012: 903.*
How is the Ejection Fraction (EF) calculated?	EF=(EDV-ESV)/EDV (EDV=End-diastolic volume, ESV=End-systolic volume) *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 514.*
What factors are associate with increased need for blood transfusion during cardiac surgery?	1. Advanced age (greater than 70 years), 2. low red cell volume due to anemia or small body size, and 3. urgent or prolonged surgery with long cardiopulmonary bypass times. *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 520.*
Define unstable angina.	Unstable angina is longer than 30 minutes in duration, exhibits ST or T-segment changes, and is new onset. This type of angina frequently is refractory to medication or rest. *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 355.*
Define stable angina.	Angina is classified as stable if the duration, frequency, or precipitating event has not changed for the last 2 months. *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 355.*

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Describe the anesthetic goals of CABG.	Ensure cerebral and myocardial protection, abolish autonomic reflexes, ensure physiologic homeostasis, and provide muscle relaxation, analgesia, and amnesia. *Longnecker DE, Newman MF, Brown DL, Zapol WM. Anesthesiology. 2nd ed. New York: McGraw-Hill; 2012: 903, 912.*
For every 1 degree C decline in core temperature, how much is the metabolic requirement for O2 reduced?	It is reduced by 8% *Miller RD, Pardo MC. Basics of Anesthesia. 6th ed. Philadelphia: Elsevier Saunders; 2011: 408.*
What are the benefits of hypothermia during CPB?	Reduced oxygen consumption, BMR is reduced, the preservation of tissues and organs, and increased myocardial protection *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 519.*
What is the normal EF?	55%-70% *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 514.*
What test is used to monitor the use of heparin during CPB?	the ACT (Activated Clotting Time) *Barash PG, Cullen BF, Stoelting RK, Cahalan MK, Stock MC, Ortega R. Clinical Anesthesia. 7th ed. Philadelphia, PA: Lippincott Williams and Wilkins; 2013: 1088.*
What is the initial dose of heparin given prior to the initiation of CPB?	300-400 units/kg. Maintenance doses are given based on the ACT values assessed during surgery. *Longnecker DE, Newman MF, Brown DL, Zapol WM. Anesthesiology. 2nd ed. New York: McGraw-Hill; 2012: 907.*
What is the desired ACT value during CPB?	greater than 400 seconds *Longnecker DE, Newman MF, Brown DL, Zapol WM. Anesthesiology. 2nd ed. New York: McGraw-Hill; 2012: 907.*
What is the purpose of the cardiopulmonary bypass (CPB) pump?	The CPB pump acts as the patient's heart and lungs in order to provide a surgeon a bloodless field and a non-beating heart. Blood is removed from the venous circulation via cannulas usually in the right atrium, or superior and/or inferior vena cava, oxygenated and CO2 elimintated, then returned to the body through an arterial cannula located in the ascending aorta. *Longnecker DE, Newman MF, Brown DL, Zapol WM. Anesthesiology. 2nd ed. New York: McGraw-Hill; 2012: 899.*

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What factors are important for the anesthetist to consider upon weaning a patient from bypass.	The aortic cross-clamp time and the pre-bypass function of the ventricle are important factors to consider. Patients with good ventricular function and short cross-clamp times usually do not require inotropic support after successful revascularization of the coronaries. Also, the patient's perfusion status, as well as body termperature and heart rhythm should be assessed. *Longnecker DE, Newman MF, Brown DL, Zapol WM. Anesthesiology. 2nd ed. New York: McGraw-Hill; 2012: 908.*
What are the two different types of oxygenators used on a CPB pump?	Membrane oxygenators and bubble oxygenators *Longnecker DE, Newman MF, Brown DL, Zapol WM. Anesthesiology. 2nd ed. New York: McGraw-Hill; 2012: 899.*
How can venous drainage to the venous reservoir of the CPB pump be improved?	By raising the height of the operating table or placing a small amount of negative pressure on the cardiotomy reservoir. *Miller RD, Pardo MC. Basics of Anesthesia. 6th ed. Philadelphia: Elsevier Saunders; 2011: 407.*
In reference to the configuration of the CPB circuit, where are membrane oxygenators positioned? Where are bubble oxygenators positioned?	Membrane oxygenators are placed after the pump because they have a higher resistance to flow. Bubble oxygenators are located before the pump because they possess reservoirs of oxygenated blood and blood does not have to be pumped through them. *Miller RD, Pardo MC. Basics of Anesthesia. 6th ed. Philadelphia: Elsevier Saunders; 2011: 406.*
What benefits are associated with the use of beta blockers during open heart surgery?	They reduce the use of oxygen through a reduction in contractility, blood pressure, and heart rate. By slowing the heart rate, the left ventricle has more time to be oxygenated due to the increased diastolic time. Beta blockers can also reduce the incidence of ventricular arrhythmias induced by catecholamines. *Longnecker DE, Newman MF, Brown DL, Zapol WM. Anesthesiology. 2nd ed. New York: McGraw-Hill; 2012: 903.*
Where would the most accurate measure of core temperature be obtained?	At the themistor of a PA catheter *Miller RD, Pardo MC. Basics of Anesthesia. 6th ed. Philadelphia: Elsevier Saunders; 2011: 330.*
What modalities may need to be considered if the heart fails to spontaneously defibrillate?	Electrical cardioversion and/or antiarrhythmic medication may be required *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 531.*
What is the drug most commonly used to reverse heparin-induced anticoagulation?	Protamine *Miller RD, Pardo MC. Basics of Anesthesia. 6th ed. Philadelphia: Elsevier Saunders; 2011: 427.*

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What device may be considered for patients that fail to wean from CPB despite optimal drug therapies and the placement of an IABP?	A Ventricular Assist Device (VAD) *Barash PG, Cullen BF, Stoelting RK, Cahalan MK, Stock MC, Ortega R. Clinical Anesthesia. 7th ed. Philadelphia, PA: Lippincott Williams and Wilkins; 2013: 1099.*
Why is the blood pressure lowered prior to cannulation of the aorta?	In order to reduce aortic wall tension and reduce the risk of aortic dissection *Longnecker DE, Newman MF, Brown DL, Zapol WM. Anesthesiology. 2nd ed. New York: McGraw-Hill; 2012: 908.*
What are the risk factors that pose the most significant risk for increased mortality following CABG?	renal failure, emergency surgery, history of cardiac surgery, >80 years of age *Longnecker DE, Newman MF, Brown DL, Zapol WM. Anesthesiology. 2nd ed. New York: McGraw-Hill; 2012: 909.*
What is the importance of monitoring glucose during cardiac surgery?	Hyperglycemia increases the risk for infection and the risk for serious neurologic injury. *Miller RD, Pardo MC. Basics of Anesthesia. 6th ed. Philadelphia: Elsevier Saunders; 2011: 410.*
Immediately prior to sternotomy what is the single most prudent action taken by the anesthesia provider?	Ventilation is interrupted (typically 15-20 seconds) to allow the lungs to deflate. This decreases the likelihood that the lung parenchyma will be cut by the sternal saw. *Longnecker DE, Newman MF, Brown DL, Zapol WM. Anesthesiology. 2nd ed. New York: McGraw-Hill; 2012: 907.*
What two valves most commonly necessitate surgical involvement?	Aortic and mitral valve *Longnecker DE, Newman MF, Brown DL, Zapol WM. Anesthesiology. 2nd ed. New York: McGraw-Hill; 2012: 909.*
What is the purpose of an intra-aortic balloon pump (IABP)?	It increases the supply of oxygen to the myocardium (diastolic augmentation) and decreases the myocardial oxygen demand (decreases the afterload) *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 548-549.*
When does inflation of the balloon of an IABP occur?	During diastole. This allows blood to be pushed into the coronary arteries. *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 548-549.*
What surgical technique would be most appropriate for a patient with an aortic arch lesion?	Hypothermic circulatory arrest. *Longnecker DE, Newman MF, Brown DL, Zapol WM. Anesthesiology. 2nd ed. New York: McGraw-Hill; 2012: 916.*

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What is the purpose of the MAZE procedure?	It is used to eliminate atrial fibrillation (AF). Patients who have undergone unsuccessful pharmacologic treatment and are at high risk for stroke are candidates for this procedure. The open heart procedure's name is derived from the serial maze-like incisions made in the atria. *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 542-543.*
What is the purpose of using miniaturized circuits during cardiopulmonary bypass?	Standard CPB circuits are associated with dilutional anemia, inflammatory reactions, hemodilution, and coagulopathy. In order to reduce these effects, miniaturized CPB circuits have been developed. *Barash PG, Cullen BF, Stoelting RK, Cahalan MK, Stock MC, Ortega R. Clinical Anesthesia. 7th ed. Philadelphia, PA: Lippincott Williams and Wilkins; 2013: 1091.*

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